

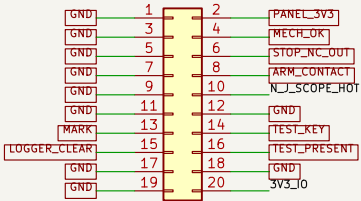
J_P12: IDC 2x10 do P12 (S1 8, PANELCORE + PANELSAFE). Piny 10/13/14/15/16 jak P03 R6 J_BP1 (N_J_SCOPE_HOT, MARK, TEST_KEY, LOGGER_CLEAR, TEST_PRESENT).

Nieparzyste GND poza 13 MARK i 15 LOGGER_CLEAR (wyjątek jak w P03 R6). 12 i 18 GND (rezerwa). 3V3_IO na 20 – z dala od PANEL_3V3 (pin 2).

PANELSAFE do P04 (wariant pełny): PANEL_3V3 2, MECH_OK 4, STOP_NC_OUT 6, ARM_CONTACT 8, TEST_KEY 14 (wspólny z P03). W LOGGER piny 4/6/8 bez odbiornika.



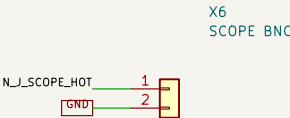
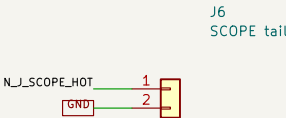
J_P12
IDC 2x10 do P12



R1 100R (decyzja P11-3): LOGGER bez P04 = OBSADZONY. Wariant z P04 = DNP (nie montować): PANEL_3V3 zasila wtedy tylko P04 R40 100R z 3V3_IO.

Obsadzony R1 przy P04 = dwa źródła (2 x 100R równolegle): znika ograniczenie prądu R40. Brak R1 bez P04 = TEST_KEY, LOGGER_CLEAR, TEST_PRESENT zawsze L.

Zwarcie PANEL_3V3-GND przy R1: 33 mA, 0.11 W w 1206 (0.25 W). Odbiorniki 3 x 10k do GND na P03 R6 (R27, R6, R8); MARK: 10k do 3V3_CORE na P03 R6.



SCOPE: N_J_SCOPE_HOT z P03 R6 przez R12 330R. BNC izolowany, odbiornik >= 1 Mohm, bez terminacji 50R. Ekran = GND.

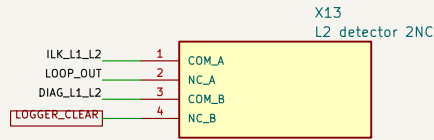
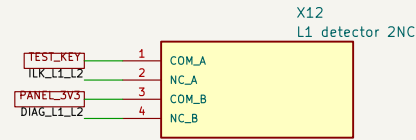
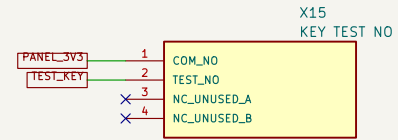


Na P11 nie ma: prądu silnika, TAPS, ISERIES, 5V_SENSOR, J1, J7, J9, J10 ani wiązek W3-W7 z R1 (P11-1, P11-4, P11-5).

Numery pinow X11...X17 sa FUNKCYJNE, nie katalogowe. P11-7: zwykle przyciski i przełączniki ze stykami ZŁOCONYMI (ok. 0.3 mA / 3.3 V), bez podświetlenia.

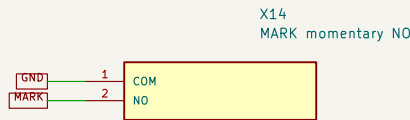
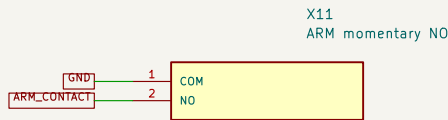
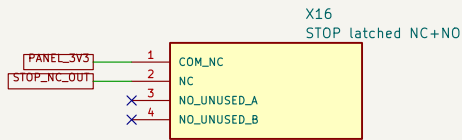
Lancuch MECH: PANEL_3V3 – kluczzyk X15 – TEST_KEY – X12 NC_A – ILK_L1_L2 – X13 NC_A – LOOP_OUT – port TEST 10 – mostek w adapterze AT – port TEST 11 – MECH_OK.

Lancuch DIAG: PANEL_3V3 – X12 NC_B – DIAG_L1_L2 – X13 NC_B – LOGGER_CLEAR. STOP: PANEL_3V3 – X16 NC – STOP_NC_OUT. TEST_PRESENT: PANEL_3V3 – X17 NO.

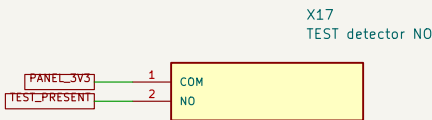
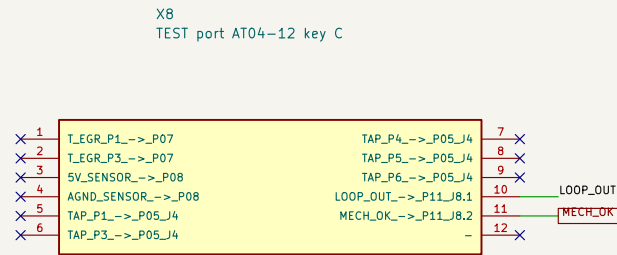


L1/L2: obie pary NC zamknięte przy pustym porcie; rozwarcie przed pierwszym stykiem elektrycznym wtyku. TEST NO zamyka dopiero po pełnym osadzeniu.

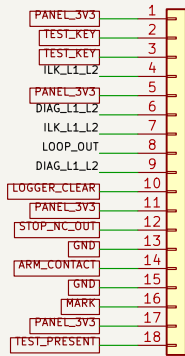
ARM i MARK zwierają do GND; podciąganie na P04 (ARM) i P03 R6 (MARK). W LOGGER ARM_CONTACT bez odbiornika.



J11
CONTACTS



J8
TEST signal



J8: tylko komory 10/11 portu TEST. Pozostałe komory portów idą przewodami wprost do P05 / P06 / P07 / P08 (arkusz PORTY, docs/PORTY.csv).

	1	2	3	4	5	6	7	8																																																																								
P11–R2 / 03 PORTY L1 / L2 / TEST (POZA P11)																																																																																
A	Porty na panelu: Amphenol AT04–12, klucz A = L1, B = L2, C = TEST (decyzja P11–2). Komory jak w R1 (adaptery AL1 / AL2 / AT bez zmian).																																																																															
Zadna komora L1 / L2 nie łączy się z P11: przewody z portu wprost do P05 J4 (TAPS, GND) i P06 J3 (ECU_P1, EGR_P1) – P11–4, P11–5.																																																																																
TEST: komory 1/2 -> P07 (T_EGR), 3/4 -> P08, 5–9 -> P05 J4; 10/11 -> P11 J8 (arkusz CONTACTS). TAPy trzech portow łącza się przy portach (docs/WIAZKI.md).																																																																																
B	X2 L1 port AT04–12 key A <table><tr><td>×</td><td>1</td><td>ECU_P1_–>_P06_J3</td><td>TAP_P6_–>_P05_J4</td><td>7</td><td>×</td></tr><tr><td>×</td><td>2</td><td>EGR_P1_–>_P06_J3</td><td>–</td><td>8</td><td>×</td></tr><tr><td>×</td><td>3</td><td>TAP_P1_–>_P05_J4</td><td>–</td><td>9</td><td>×</td></tr><tr><td>×</td><td>4</td><td>TAP_P3_–>_P05_J4</td><td>–</td><td>10</td><td>×</td></tr><tr><td>×</td><td>5</td><td>TAP_P4_–>_P05_J4</td><td>–</td><td>11</td><td>×</td></tr><tr><td>×</td><td>6</td><td>TAP_P5_–>_P05_J4</td><td>GND_–>_P05_J4</td><td>12</td><td>×</td></tr></table> X3 L2 port AT04–12 key B <table><tr><td>×</td><td>1</td><td>TAP_P1_–>_P05_J4</td><td>–</td><td>7</td><td>×</td></tr><tr><td>×</td><td>2</td><td>TAP_P3_–>_P05_J4</td><td>–</td><td>8</td><td>×</td></tr><tr><td>×</td><td>3</td><td>TAP_P4_–>_P05_J4</td><td>–</td><td>9</td><td>×</td></tr><tr><td>×</td><td>4</td><td>TAP_P5_–>_P05_J4</td><td>–</td><td>10</td><td>×</td></tr><tr><td>×</td><td>5</td><td>TAP_P6_–>_P05_J4</td><td>–</td><td>11</td><td>×</td></tr><tr><td>×</td><td>6</td><td>–</td><td>GND_–>_P05_J4</td><td>12</td><td>×</td></tr></table>								×	1	ECU_P1_–>_P06_J3	TAP_P6_–>_P05_J4	7	×	×	2	EGR_P1_–>_P06_J3	–	8	×	×	3	TAP_P1_–>_P05_J4	–	9	×	×	4	TAP_P3_–>_P05_J4	–	10	×	×	5	TAP_P4_–>_P05_J4	–	11	×	×	6	TAP_P5_–>_P05_J4	GND_–>_P05_J4	12	×	×	1	TAP_P1_–>_P05_J4	–	7	×	×	2	TAP_P3_–>_P05_J4	–	8	×	×	3	TAP_P4_–>_P05_J4	–	9	×	×	4	TAP_P5_–>_P05_J4	–	10	×	×	5	TAP_P6_–>_P05_J4	–	11	×	×	6	–	GND_–>_P05_J4	12	×
×	1	ECU_P1_–>_P06_J3	TAP_P6_–>_P05_J4	7	×																																																																											
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×	6	–	GND_–>_P05_J4	12	×																																																																											
C																																																																																
D	Dopuszczony jeden adapter naraz. P11 nie wybiera źródła TAP; wspólne TAPy wymagają procedury lub przesłony mechanicznej portow (jak R1).																																																																															
Prad silnika (do 6 A, 10 A w probie biernej): przewody >= 1.0 mm2 z portu wprost do P06 / P07, styki AT 13 A. Pomiar cieplny całej petli w odbiorze.																																																																																
E																																																																																
F	1	2	3	4	5	6	7	8																																																																								

Panel contact logic and P12 ribbon; format S1 (decisions P11–1..P11–7, 4.10.2026); schematics

Prototype / schematic + PCB review

Sheet: /PORTY/
File: PORTY.kicad_sch

Title: EGRLab P11 PANEL

Size: A3

Date: 2026–10–04

Rev: P11–R2

KiCad E.D.A. 10.0.6

Id: 3/3